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# Effects of cell lysis in gas yield and sludge stabilization by combined process of thermophilic aerobic and anaerobic digestion

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**Abstract**

A novel process of combining thermophilic (55 °C) aerobic digestion as pretreatment with mesophilic (<35 °C) anaerobic digestion process (TAD-MAD) was designed to stabilize sludge and produce CH4 biogas. TAD-MAD processes can yield energy-rich biogas, save aeration energy, short retention time, which is prior to single TAD or MAD process. TAD pretreatment play an important role in decomposing cell of sludges and enhancing the hydrolysis process. Other two pretreatment of Alkali and Ultrasonic following MAD process were investigated in this paper to compare with effect of TAD-MAD process on sludge stabilization and biogas production. The results showed that VSS removal rates for three processes of TAD-MAD, Alkali-MAD and Ultrasonication-MAD could achieve 40% at 10d or 11d, which can meet the requirements of the GB18918-2002 guidelines. soluble chemical oxygen demand (SCOD) and volatile fatty acid (VFA) in the supernatant could increase to the highest values at initial 4d for three different processes, and subsequently decline during the latter part of digestion. For TAD-MAD process, SCOD and VFA could rapidly reach to the highest values 16000 mg/L and 7222 mg/L at initial 2d respectively, which achieved more significant effect on cell lysis. All three pretreatments improved biogas production, but TAD-MAD process could produce the most amounts of CH4 (481mL CH4 g−1 VSS) and cumulative biogas production of 12496mL at end of digestion. pH between 7.04 and 7.92 in three systems could maintain in neutral and subalkaline values. Alkalinities of sludge in three systems could meet need for methanogenic population inside the anaerobic digester. In conclusion, TAD-MAD has similar VSS removal rate with other two processes, but could produce more CH4 and biogas yield than other two processes.

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*Keywords: Pretreatment; thermophilic aerobic digestion; alkali; ultrasonication; cell lysis; biogas;*

## 1. Induction

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Sewage sludge disposal has been a puzzle for most of Wastewater Treatment Plants (WWTPs)[1]. Biological treatment methods of anaerobic and aerobic digestion processes are commonly used to achieve sludge stabilization. Among these processes, autothermal thermophilic aerobic digestion (ATAD) has been paid significant attention in sludge stabilization because of its produce Class A biosolids [1-4].

However, process of ATAD can't produce biogas of hydrogen and methane [5,6]. Although anaerobic digestion process(AD) can provide useful methane production, and doesn't require an oxygen supply[7], it clearly has limitations in terms of long retention times to achieve biostabilization and low overall degradation efficiency of the organic matter. Actually, the rate-limiting step in WAS digestion is the hydrolysis of organic matter, or cell walls lysis[8]. Therefore, a novel combined processes of thermophilic aerobic digestion and mesophilic anaerobic digestion (TAD-MAD) in two-stage digester has been investigated by bath mode. The processes TAD-MAD can yield energy-rich biogas, save aeration energy, short retention time to stabilize the sludge.

Therefore, TAD as pretreatment process need to be investigated whether to better lyse the cells in waste activated sludge(WAS) to readily biodegradable low-molecular-weight compounds. Many different pretreatment technologies have been studied to overcome these limitations, such as mechanical[9], ultrasonic[10], alkaline[9,11], and enzymatic[12] methods could disrupt the cell walls and increase WAS biodegradability by enhancing the hydrolysis process.

This work focus on combination of co-treatments with mesophilic anaerobic digestion in WAS system. The sludges were treated by TAD-MAD, in comparison with other two pretreatment processes of alkali and ultrasound to determine the effect of cell lysis. Based on the obtained results of optimum operation parameters and fundamental theories of aerobic and anaerobic digestions, batch experiments are performed. This study investigates the factors of VS removal, release of intercellular organic matters and volatile acid, production of biogas and methane. This work aims to evaluate the effectiveness of TAD-MAD and provide recommendations for its large-scale application.

## 2. Materials and methods

### *2.1 Sample of sludge*

Sewage sludge was sampled from a municipal wastewater treatment plant in Changzhou, China in which an anaerobic-anoxic–oxic process was applied to remove biological nitrogen and phosphorus. The sampling sludge, which had originally been stored for about 24 h in a settling tank, was centrifuged at 3500 rpm for 5 min to obtain a total suspended solid (TSS) level between 5% and 6% solids.

### *2.2 Start-up of tree different pretreatment process*

Three combined processes are MAD digestion connecting with three different processes alkaline, ultrasonic and TAD as pretreatment, respectively. After the pretreatment process was initiated, MAD process was operated. In Alkali-MAD process, the sludge was firstly treated by alkali (pH of 12, NaOH) for 0.5h under room temperature at 24°C during process of alkaline pretreatment. In Ultrasonication-MAD process, the sludge was treated by ultrasonication at 20kHz and 60w (FS-300, Shengxi Ultrasonic Instrument Co. Ld., Shanghai ) for 0.5h during ultrasonic treatment process. In TAD-MAD process system, the sludge was aerobically operated for two days in batch mode at 55°C for TAD process as pretreatment. The sludges, treated by three different pretreatment processes, were inputted in MAD digester to be anaerobically digested for 18d at 35°C, respectively.

### *2.3 Experimental facilities*

The process of TAD-MAD is consisted of two digesters of TAD and MAD, which were two cylindrical glass digesters in a water bath shaker to maintain a specific digestion temperature. MAD digester is 3L effective volume with 40mm diameter and 65mm height. A gas collection bottle is connected on top of MAD digester by an effluent gas pipe for the collection of effluent gas. Effective volume of TAD digester is 2 L, and 30mm diameter and 50mm height. Air was supplied through a microporous diffuser in TAD digester with a flow rate of 0.1 L $min^{-1}$,[13] .

### *2.4 Analysis method*

Sludge samples were collected every 48 h to determine TSs, and VSs as sludge stabilization indicators, as well as others two of soluble chemical oxygen demand (SCOD) and volatile fatty acid (VFA) in the supernatant of the sludge sample. The pH was not regulated and used as an indicator to reflect anaerobic digestion stabilization during the entire digestion process. The pH and alkalinity of sludge were monitored every 24 h. Total gas volume and methane yield were collected and analysed.

Analyses of TSs and VSs of the sludge samples were performed according to the standard methods[14]. The supernatant of the sludge samples can be obtained when the samples were previously centrifuged at 12,000 g for 15 min, and the supernatant was filtered through a 0.45 μm pre-washed membrane filter (AdvantecGC50, glass fiber). VFA and SCOD were measured by the standard reflux trimetric method using the distillation-boric acid absorption-titration method[15]. The pH of sludge were measured by a pH meter (pH/mV/°C Meter EUTECH Instruments, Model pH 11 series). All these indicators, except for pH and alkalinity were determined in triplicate, and the standard deviations of all analyses were less than 5%.

## 3. Results and Discussion

### *3.1 The effect on sludge stabilization*

The removal rates of VS ($\eta_{vs}$) were a significant performance indicator to embody the level of sludge stabilization. The results of removal rate of VSS were presented in Fig. 1 for all experiments at different pretreatment processes throughout 18 d of digestion.

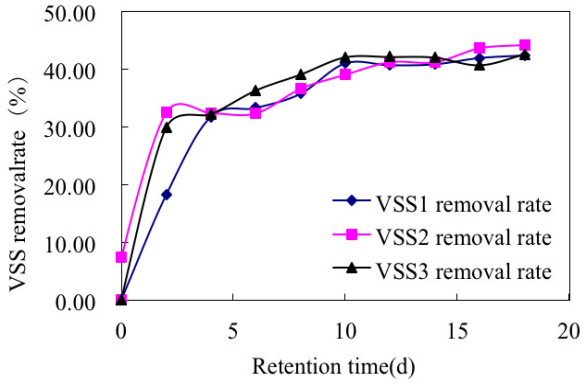



Fig. 1 Variation of VSS removal rate of three different processes

VSS1 removal rate：TAD-MAD process
VSS2 removal rate：Alkali-MAD process
VSS3 removal rate：Ultrasonication-MAD process

As showed in Fig. 1, the VSS removal efficiency for three different processes of TAD-MAD, Alkali-MAD and Ultrasonication -MAD exhibited a rapid increase to 32% or so at 4d followed by a gradual increase afterwards, and reached 41.0%, 41.2% and 42% at 10d, 12d and 10d, respectively. The results showed that the VSS removal rate for three processes with different pretreatment could achieve 40% or more, which can meet the requirements of the GB18918-2002 guidelines (i.e., the degradation of organic matter must exceed 40% when digestion is applied to stabilize the sewage sludge from municipal WWTPs) [15] at 10 or 11d, respectively. In comparison, the VS removal rate for Ultrasonication-MAD process presented slightly higher values than other two processes from 5d to 10d. However, three processes had same VS removal rates at the end of digestion in total. The results indicated that TAD pretreatment of 2 d can reach the same VS removal rates with other two pretreatment processes of alkaline and ultrasonication after sludges were all anaerobically digested by MAD process.

In general, sludge digested by MAD process could achieve stabilization after more than one month. The results above suggested that TAD pretreatment could destroy the cell walls of sludge and promoted anaerobic biological reaction velocity to short the retention time of sludge stabilization in WAS system. Other two processes of alkaline and ultrasound could play an efficient roll in increasing VSS removal rates.

### *3.2 The effect of cell lysis*

#### *3.2.1 Change of soluble COD in the supernatant of sludge*

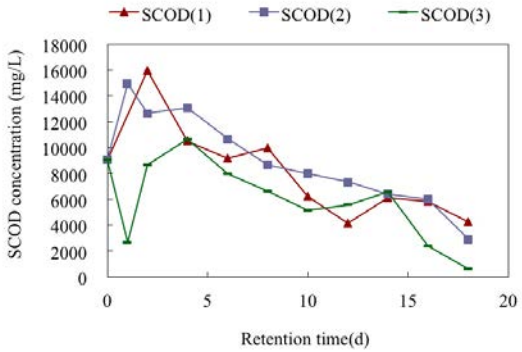



Fig. 2 Variation of SCOD of three different processes

SCOD(1)：SCOD in TAD-MAD process

SCOD(2)：SCOD in Alkali-MAD process

SCOD(3)：SCOD in Ultrasonication-MAD process

The results of soluble COD (SCOD) in the supernatant for three different processes are shown in Figs. 2. As shown in Fig. 2, SCOD for two processes of TAD-MAD and Alkali-MAD rapidly increased to the highest values of 16000 and 14933 mg/L at 2 d and 1d, respectively. The results suggested that cell walls were disintergrated to cause a large number of intracellular compounds released in the supernatant when sludges treated by TAD and alkali. However, SCOD for Ultrasonication-MAD process presented lower value of 2667mg/L at 1d, and rose at the highest value of 10667 mg/L less than that of SCOD in TAD-MAD and Alkali-MAD. Evidence from the literatures of sludge reduction mechanisms suggests that abundant intercellular organic materials, including some thermostable enzymes, were released to the supernatant due to enhanced endogenous decay and cell walls lysis[16,17], thereby contributing to the sharp increase in SCOD concentration. Therefore, effect of cell lysis for TAD was more efficient than that of ultrasonication, and similar with that of alkali.

After maintaining high values for 1 d, SCODs began to decline sharply during the latter part of digestion, indicating that SCODs were utilized to produce biogas afterward in MAD system. Methane production was benefit from high SCOD concentration. As a result, TAD-MAD system is prior to other two processes in cell lysis action.

#### *3.2.2 Change of VFA in the supernatant of sludge*

The VFA concentrations in the supernatant of sludge in three different processes were investigated. And the results were shown in Fig.3.

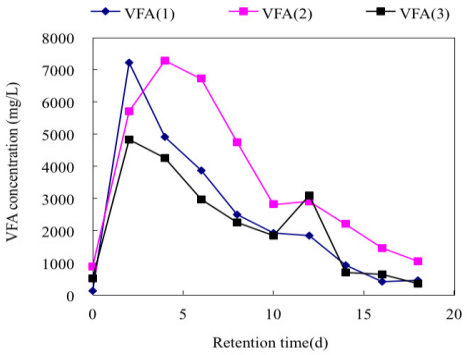



Fig. 3 Variation of VFA of three different processes

VFA(1)：VFA in TAD-MAD process

VFA(2)：VFA in Alkali-MAD process

VFA(3)：VFA in Ultrasonication –MAD process

The highest concentrations of VFA in TAD-MAD and Alkali-MAD processes occurred on day 2 and day 4, respectively (Fig. 3). VFA concentrations reached a maximum value of 7222 at 2 d and 7282 mg/L at 4d in the TAD-MAD and Alkali-MAD system, respectively. However, a maximum value of VFA at 4828mg/L in Ultrasonication-MAD system was comparatively lower than that of other two systems. The results indicated that two pretreatment processes of TAD and Alkali efficiently led to more cell lysis than ultrasonic treatment. Pretreatment aims at intensification by enhancing the hydrolysis process in an additional stage and transforming intercellular organic matters to VFA. Ultrasonic treatment acts to mechanically disrupt the cell structure and floc matrix. However, limited frequency and voltage did not produce cell membrane breakage, unless the disintegration degree was over 40%[18].

With increasing retention time, VFA sharply declined up to low values at the end of digestion in three different systems of TAD-MAD, Alkali-MAD and Ultrasonication MAD. The initial pretreatment of sludge resulted in high VFA in the liquid phase because of the cell walls degraded and intercellular matters microbial fermentation process. The initial high VFA contents were advantageous to the following MAD digestion process and biogas production because VFA was used easily, thereby causing its rapid decrease.

### *3.3 Efficiency of gas production*

Pretreatment process generally leads to sludge floc disintegration and microorganisms lyses, and increase potential biogas production. Therefore, the methane production and total gas yields should be investigated for three different processes to display the effect of cell lysis. The results were shown in Fig.4 and Fig.5 .

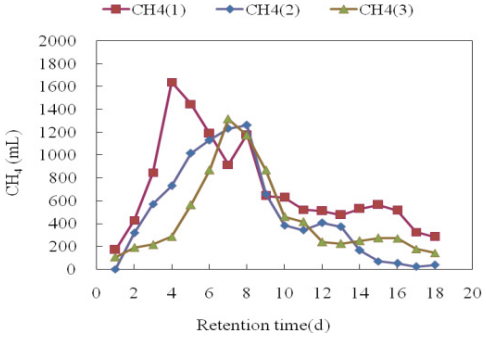



Fig. 4 Variation of methane yield of three different processes

$CH_4$(1)：$CH_4$ yield in TAD-MAD process
$CH_4$(2)：$CH_4$ yield in Alkali-MAD process
$CH_4$(3)：$CH_4$ yield in Ultrasonication-MAD process

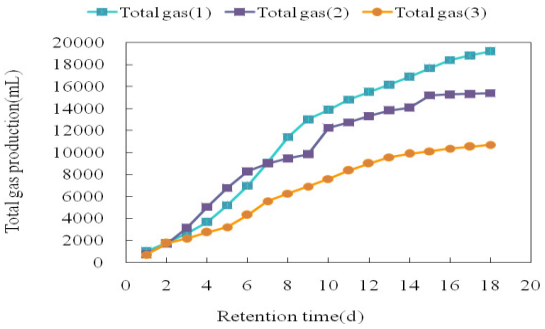



Fig. 5 Variation of methane yield of three different processes
Total gas(1)：Cumulative gas yield in TAD-MAD process
Total gas(2)：Cumulative gas yield in Alkali-MAD process
Total gas(3)：Cumulative gas yield in Ultrasonication-MAD process

From Figs. 4, The methane volumes in three different processes of TAD-MAD, Alkali-MAD and Ultrasonication-MAD increased continuously in the first 4, 8, and 7 d, reaching the maximum values of 1634, 1284, and 1315 mg/L, respectively, indicating that they have reached maximum values. After reaching maximum volumes, $CH_4$ production in three different processes sharply declined during latter digestion period, which implied that degradable organic matter such as VFA were used in mesophilic anaerobic digestion process.

In addition, total biogas productions in three different processes were cumulated gas yield, shown in Fig 5. Total biogas yields similarly trend to increase rapidly during total retention time. In comparison, total biogas yields in TAD-MAD system rose speedily more than that of other two systems, and cumulative biogas production reached12496mL at end of digestion. Gas yield of sludge treated by Ultrasonic-MAD process presented the lowest values among three processes.

In terms of biogas yields, TAD biological treatment was the most efficient. All three pretreatment processes improved biogas production. For Ultrasonication-MAD process, this enhancement was the lowest value of 246mL $CH_4$ for per gram removal VSS compared to Alkali-MAD (319mL $CH_4$ $g^{-1}$ VSS) and TAD-MAD (481mL $CH_4$ $g^{-1}$ VSS). According to others studies, ultrasonic treatment provided minimal solubilization of sludge and particle size reduction, but improved biodegradability of the particulate fraction[19]. In this paper, ultrasonic treatment didn't present the highest $CH_4$ production during Ultrasonication-MAD process.

### *3.4 Variation of pH and alkalinity*

#### *3.4.1 Alkalinity of sludge*

Variations of alkalinity for three processes were shown in Fig.6. Anaerobic treatment generally calls for a neutral condition for a methane production process. A large quantity of VFA released in the supernatant during anaerobic digestion caused alkalinity decrease and a risk to trend acid environment.

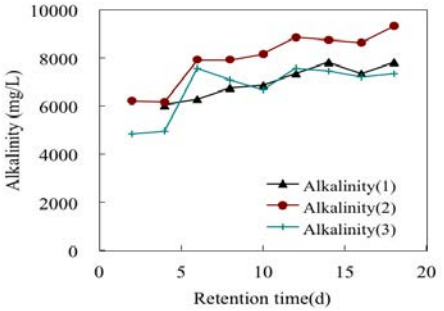



Fig. 6 Variation of alkalinity of three different processes

Alkalinity(1)：Alkalinity of sludge in TAD-MAD process

Alkalinity (2)：Alkalinity of sludge in Alkali-MAD process

Alkalinity (3)：Alkalinity of sludge in Ultrasonication-MAD process

According to Fig.6, the alkalinities of sludge in three different systems gradually rise with retention time prolonging. In total, the alkalinities of TAD-MAD and Ultrasonication-MAD were measured between 6024~8032 mg/L and 4842~7559 mg/L during whole anaerobic digestion period respectively, which were lower than that of Alkali-MAD. Due to alkalinity sodium hydroxide agent dosed in the Alkali-MAD system, alkalinities were evidently shown higher values. In the initial study period, gas production for the control was low due to the low alkalinity in TAD-MAD system. This is consistent with the VFA data that indicated that the system may not have been entirely acclimated. Gas production for TAD-MAD was much steadier in subsequent period and this reflects the acclimation of the organisms in general.

Alkalinities of three different processes were sufficient and result in pH more than 7.0 during subsequent anaerobic digestion, as shown in Fig.7. The results suggested that values of alkalinities could meet need for methanogenic population inside the anaerobic digester to produce methane gas, though VFA in the supernatant were observed high values in initial anaerobic period. Proper alkalinities for anaerobic digestion process play an important role in sludge stabilization and gas production.

### *3.4.2 pH of sludge*

As to anaerobic digestion process, a lower pH will result in the growth of filamentous bacteria and a high pH results in buildup of ammonia[20]. Anaerobic treatment under acidic and alkaline conditions may play an important future role when anaerobic digeston treatment processes in industries call for a higher tolerance for extreme conditions. Therefore, pHs of sludge in three different processes were analyzed to observe whether the operations of three different MAD phrases could be normal. The results of pHs were shown in Fig.7.

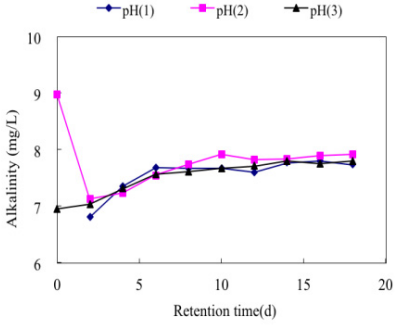



Fig.7 Variation of pH of three different processes

pH(1)：pH of sludge in TAD-MAD process
pH (2)：pH of sludge in Alkali-MAD process
pH (3)：pH of sludge in Ultrasonication-MAD process

Anaerobic Digestion is generally operated under neutral pH conditions (pH 6.5-7.8). The potential risk would be existed because the presence of VFA in sludge could cause toxicity under low pH conditions[21]. In Fig.7, pH of sludge in two processes of TAD-MAD and Ultrasonication-MAD all presented the values between 7.04 and 7.92 from 2d. Although initial pH in Alkali-MAD system were shown maximum value at 8.98, including pH at 6.82 in TAD-MAD system and 6.96 in Ultrasonication-MAD system, pH could maintain in neutral and subalkaline values between 7.04 and 7.92.

The results indicated that variation of pHs for three processes during digestion period didn't cause a large quantity of hazardous effect on normal anaerobic digestion and gas production. The methanogenic activity will slow considerably at a pH less than 6.3 and higher than 7.8 and this will inhibit biogas production[22].

## 4. Conclusion

(1)VSS removal rates for three processes of TAD-MAD, Alkali-MAD and Ultrasonication-MAD can achieve 40% at 10d or 11d, which can meet the requirements of the GB18918-2002 guidelines.

(2) SCOD and VFA in the supernatant could rapidly increase to the highest values at initial 4d for three different processes, and subsequently decline during the latter part of digestion, indicating that pretreatment could promote cell lysis and have advantage of producing biogas afterward in MAD system. In comparison, TAD pretreatment process is most efficient in SCOD and VFA production among three processes.

(3) All three pretreatments improved biogas production, but TAD biological treatment was the most efficient.

(4) TAD-MAD has similar VSS removal rate with other two processes, but could produce more $CH_4$ and biogas yield than other two processes.

(5) Alkalinities of sludge in three systems could meet need for methanogenic population inside the anaerobic digester to produce methane gas. pH could maintain in neutral and subalkaline values between 7.04 and 7.92.

## Acknowledgements

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## References

1. Ma, S Y, Tang, J G, and Chen, B L. Sludge disposal and utilization in EU. *China Water and Wastewater*. 2006, 22:102–105 (in Chinese).
2. USEPA. Autothermal Thermophilic Aerobic Digestion of Municipal Wastewater Sludge Report, EPA/625/10-90/007. Environmental Protection Agency, Washington, DC. 1990.
3. USEPA. Standards for the Use or Disposal of Sewage Sludge, United States Environmental Protection Agency, Washington. 1993.
4. Juteau P, Tremblay D, Ould-Moulaye C B, Bisaillon J G, Beaudet R. Review of the use of aerobic thermophilic bioprocesses for the treatment of swine waste, *Livestock Science*, 2006, 102: 187-196.
5. Ugwuanyi J O. Aerobic Thermophilic Digestion of Model Agricultural Wastes. Ph.D. Thesis. University of Strathclyde, Glasgow, UK.1999
6. Cheng J H, Ji Y H, Kong F, Chen X. Combined Mesophilic Anaerobic and Thermophilic Aerobic Digestion Process: Effect on Sludge Degradation and Variation of Sludge Property, *Applied Biochemistry and Biotechnology*, 2013, 171: 1701–1714.
7. Yang X Y, Wang X, Wang L. Transferring of components and energy output in industrial sewage sludge disposal by thermal pretreatment and two-phase anaerobic process. *Bioresource Technology*, 2010, 101(8): 2580-2584
8. Li Y Y, Noike T. Upgrading of anaerobic digestion of waste activated sludge by thermal pretreatment. *Water Science and Technology*, 1992, 26(3-4): 857-866.
9. Muller C D, Abu-Orf M, Novak J T. Application of mechanical shear in and internal- recycle for the enhancement of mesophilic anaerobic digestion. *Water Environment Research*, 2007, 79(3): 297-304.
10. Jin Y, Li H, Mahar R B, Wang Z, Nie Y. Combined alkaline and ultrasonic pretreatment of sludge before aerobic digestion. *Journal of Environmental Sciences*, 2009, 21(3): 279-284.
11. Shao L M, Wand G Z, Xu H Z C, Yu G H, He P J. Effects of ultrasonic pretreatment on sludge dewaterability and extracellular polymeric substances distribution in mesophilic anaerobic digestion. *Journal of Environmental Science*, 2010, 22(3): 474-480.

12. Ayol A, Filibeli A, Sir D, Kuzyaka E. Aerobic and anaerobic bioprocessing of activated sludge: Floc disintegration by enzymes. *Journal of Environmental Science and Health, Part A Toxic/Hazardous Substances and Environmental Engineering*, 2008, 43(13): 1528-1535.
13. Cheng J H, Zhou Q F, Zhu N W. Study of oxidation-reduction potential (ORP) on autothermal thermophilic aerobic digestion process(in China). *Res. Environ. Sci.* 2009, 22: 484-489.
14. APHA, AWWA, WEF, Standard Methods for the Examination of Water and Wastewater, 21st ed., American Public Health Association/American Water Works Association/Water Environment Federation, Washington USA, 2005.
15. SEPAC, GB18918-2002. Discharge Standards for Pollutants from Municipal Wastewater Treatment Plants, *China Environmental Science Press*, Beijing, China. 2002.
16. Van Loosdrecht M C M, Henze M. Maintenance, endogenous respiration, lysis, decay and predation. *Water Sci. Technol*, 1999, 39: 107-117.
17. Liu S G, Song F Y, Zhu N W, Yuan H P, Cheng J H. Chemical and microbial changes during autothermal thermophilic aerobic digestion (ATAD) of sewage sludge, *Bioresour. Technol.* 2010,101: 9438–9444.
18. Show K Y, Mao T, Lee D J, Optimisation of sludge disruption by sonication, *Water Res.* 2007, 41 (20) : 4741–4747.
19. Bougrier C, Delgenes J P, Carrere H. Combination of thermal treatments and anaerobic digestion to reduce sewage sludge quantity and improve biogas yield, Process Saf. *Environ. Protect.* 2006, 84 (B4): 280–284.
20. Grady C P, Leslie, Daigger, Glen T., Lim H C, Biological Wastewater Treatment, Marcel Dekker, New York, 1999.
21. van Lier J B, Tilche A, Ahring, B.K., Acarie H, Moletta R, Dohanyos M, Hulshoff Pol L W, Lens P, and Verstraete W. New perspectives in anaerobic digestion. *Water Sci Technol.* 2001, 43: 1-18 .
22. Leitao R C, van Haandel A C, Zeeman G, Lettinga G. The effects of operational and environmental variations on anaerobic wastewater treatment systems: A review. *Bioresource Technol.* 2006, 97: 1105–1118.